

Periodic Functions

Many real-world phenomena occur in cycles.

Some graphs are periodic. Periodic graphs are formed by cycles, or regular intervals of repeating patterns.

Analyzing a Periodic Function

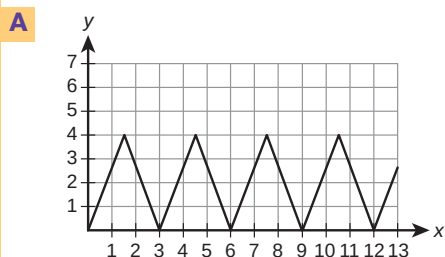
Definitions

In a periodic function, the **period** is the length of one cycle. The **amplitude** is half the difference between the maximum and minimum output values.

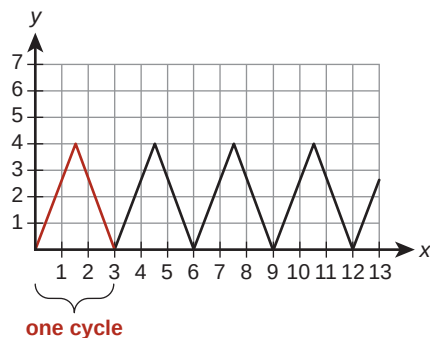
► **Think About It** A cycle is the shortest complete repeating pattern in a graph.

EXAMPLE 1

Identify the period and amplitude of the function.



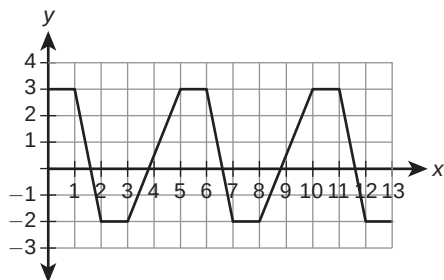
SOLUTION



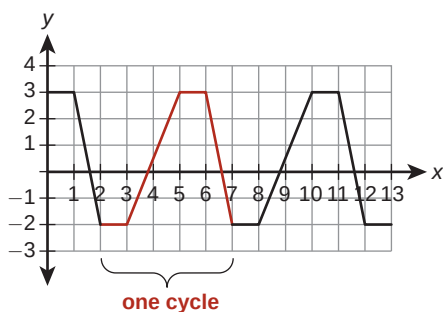
The highlighted cycle extends from $x = 0$ to $x = 3$. The period is $3 - 0$, or 3. The maximum y -value is 4 and the minimum y -value is 0. The amplitude is

$$\frac{4 - 0}{2} = 2.$$

B



SOLUTION



The highlighted cycle extends from $x = 2$ to $x = 7$. The period is $7 - 2$, or 5. The maximum y -value is 3 and the minimum y -value is -2 . The amplitude is

$$\frac{3 - (-2)}{2} = \frac{5}{2} = 2.5. \quad \blacksquare$$

Determining the Period of a Data Set

EXAMPLE 2

Determine whether the data set is approximately periodic.

If so, estimate the period and the amplitude.

The table shows the times and heights of the tides for Barren Island, in the Chesapeake Bay, during a given time span.

Time (h)	5.76	11.76	18.01	24.56	30.56	36.68	42.68	49.25
Height (ft)	1.4	0.4	1.6	0.3	1.5	0.4	1.6	0.3

Tide High and Low, Inc. 2013

SOLUTION

Look for patterns. The tides alternate between high tides at about 1.5 ft and low tides at about 0.35 ft. The times are all approximately 6.25 h apart.

The data set is periodic. The period is about $6.25 \text{ h} + 6.25 \text{ h} = 12.5 \text{ h}$.

The amplitude is about $\frac{1.5 - 0.3}{2} \text{ ft} = \frac{1.2}{2} \text{ ft} = 0.6 \text{ ft}$. ■

Graphing a Trigonometric Function

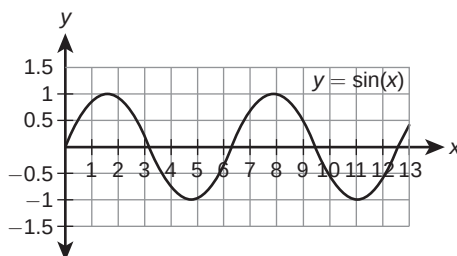
A trigonometric function, or trig function, has a trigonometric ratio in its function rule. Trigonometric functions are periodic functions.

You can graph a trigonometric function on a graphing calculator or by using an online tool.

Analyzing a Trigonometric Function

EXAMPLE 3

Identify the period and amplitude of the sine function.



SOLUTION

One cycle extends from $x = 0$ to $x \approx 6.25$. The period is about 6.25. The maximum y -value is 1 and the minimum y -value is -1 .

The amplitude is $\frac{1 - (-1)}{2} = \frac{2}{2} = 1$. ■

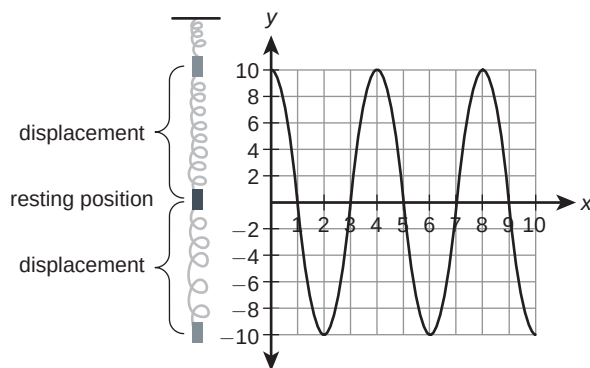
► **Think About It** Some graphing tools will show the x -values in terms of π .

The exact period of the sine function in Example 3 is 2π . The value of π is approximately 3.14, so the value of 2π is approximately 6.28.

Application: Simple Harmonic Motion

The motion created by an object shifting back and forth by regular amounts is an example of **simple harmonic motion**. Simple harmonic motion can be modeled by sine and cosine functions, called **sinusoids**.

Imagine a weight on the end of a spring as the spring is compressed upward and then released. The weight would spring down below its initial, or resting, position, and then up above it. This would continue indefinitely in the absence of friction.



The graph shows the height y , in centimeters, of the weight, at a given number of seconds, x .

Compare the graph to the table.

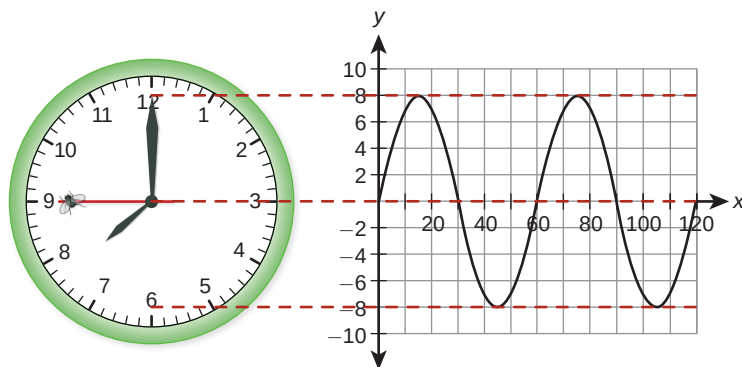
Time (s)	0	1	2	3	4	5	6	7	8
Displacement (cm)	10	0	-10	0	10	0	-10	0	10

The weight's displacement at any moment is its distance from its resting position. Its maximum displacement is the graph's amplitude: 10 cm. From its resting position, the time it takes the weight to bounce one direction, then the other, and then return to its resting position is the period: 4 s.

The **frequency** of a periodic graph is the reciprocal of its period and gives the number of cycles in one unit of time. For the spring, the frequency of $\frac{1}{4}$ indicates that the weight completes one-fourth of a cycle in 1 s.

Application: Circular Motion

Suppose a fly lands on the end of the second hand of a clock at 7:45. The graph of its vertical movement for the next 2 min would form a periodic function because it would move up and down in regular intervals.



Compare the graph to the table.

Time (s)	0	15	30	45	60	75	90	105	120
Height (in.)	0	8	0	-8	0	8	0	-8	0

The length of the second hand is the graph's amplitude: 8 in. The time it takes for the fly to travel around the clock and return to its original position is the period: 60 s.

► Q&A

Q How many degrees does the fly turn per second?

A $6^\circ/\text{s}$